Recursive Integer Spectra for Exact Deterministic Moment Prefixes

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July 2026

Abstract

The isolated packets of RH-315 can be iterated because the order-d packet does not alter any lower moment. We obtain a finite conjugate-closed spectrum inside the modulus cap whose power sums match any prescribed finite real prefix exactly. All multiplicities are integers, so the construction is a finite normal matrix rather than a weighted measure. It remains synthetic and is not identified with the actual noisy operator.

1 Triangular construction

Fix $q > 0$ and real moments $a_1, \ldots, a_N$. Begin with the empty multiset. After steps $1, \ldots, d-1$, let

$$w_d = a_d - \sum_{\mu \in \mathcal{S}_{d-1}} \mu^d.$$

If $w_d \neq 0$, choose

$$L_d \geq |w_d|/(dq^d)$$

and adjoin the packet $\mathcal{P}_d(w_d, L_d)$ of RH-315.

Theorem 1.1 (exact spectral prefix)**.** *The final multiset $\mathcal{S}_N$ is finite, conjugate closed, and contained in $|\mu| \leq q$. For every $1 \leq n \leq N$,*

$$\sum_{\mu \in \mathcal{S}_N} \mu^n = a_n.$$

Thus $\mathcal{S}_N$ is the spectrum of a finite normal matrix with integer multiplicities and the exact prescribed power-sum prefix.

Proof. At step d, the new packet has moment w_d at order d and zero moments at all lower orders. Induction therefore fixes the dth moment without disturbing the preceding ones. The radius criterion and conjugate closure hold packet by packet. A diagonal matrix with these entries is normal. □

2 Application to the deterministic anchor

Taking a_n to be the RH-263 all-order deterministic numerator anchors gives an exact finite spectral realization for every prefix. The construction is not fitted from the RH-253 table; that table is unnecessary once the all-order anchors are known.

The reproducibility experiment reads the archived RH-263 anchor rows through order eight and runs the packet recursion in floating arithmetic. This is a finite implementation check of the exact theorem, not evidence for an all-order noisy spectral fit.

Remark 2.1. Finite normal realizability does not prove that the actual noisy complement has these eigenvalues, this rank, or this moment transport. Gates A–E remain false/open.